

Assignment 3

Files Manipulation

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Assignment

Tee system call

In programming, especially when working with files, you may encounter the need to duplicate data to multiple destinations. A common scenario is application logging, where you want to view logs in real-time on the terminal and also save them to a file for later analysis. The `tee` function provides a convenient way to achieve this.

Implement the `tee` function. It reads the standard input and write it back both to:

1. the standard output
2. the file taken in parameter

```
int tee(char *output_file);
```

The standard input is represented by the file descriptor `0`. You can also use the macro `STDIN_FILENO`. The same applies for the standard output, where the macro `STDOUT_FILENO` is equal to `1`.

The `output_file` **must** be created with permissions `rw-r--r--`.

The function shall return `0` in case of a success, and `-1` in case of an error.

Read the manual

This function is inspired by the command `tee`. Check the manual: `man tee(1)`.

Reverse the bytes of a file

Reverse the content of `input_file` byte by byte. You are expected to use `lseek` to achieve this. Write your output to `output_file`.

```
int reverse_file(char *input_file, char *output_file);
```

The `output_file` **must** be created with read/write permissions.

The function shall return `0` in case of a success, and `-1` in case of an error.

Optimize your code

Optimize `reverse_file`.

This last function is deliberately harder than the rest of this assignment. Calling read/write on every byte can be very inefficient. One way to optimize the code is to use a temporary buffer to work on a small part of the file at a time.

```
int reverse_file_optimized(char *input_file, char *output_file);
```

You will need to be careful not using too much resources in terms of:

1. memory
2. execution time

If the program is taking too much time or too much memory, it will crash.

Remark

If you end up with this kind of output, it means that the program has been killed because it took too long to execute.

```
Test 5: Bigfile reverse test
Configuration error in the test case: the output is not definedProgram timeout
Incorrect exit code. Expected 0, found -2147483648
Incorrect program output
--- Input ---

--- Program output ---
```